

**Experiential Learning
Project Assessment-I**
on
“File Directory Structure Simulator”
under the course
Python Programming Language (N-PCCCM304P)



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Session: 2026-27 (ODD)
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ROADMAP

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Problem Statement

1

Manual Systems Are Slow

Processing marks for a large batch of students by hand is time-consuming and prone to human error.

2

Inconsistent Calculations

Manually working out totals and percentages for hundreds of students leads to inconsistent, error-prone results.

3

Tedious Grade & Rank Assignment

Assigning grades and ranking students by hand becomes impractical as class size grows.

4

Need for Printable Marksheets

Students and faculty need accurate, ready-to-print marksheets generated instantly, not compiled by hand.

Python Solution: Lists / dictionaries store student records and subject-wise marks, then automatically compute totals, percentages, grades, and ranks, producing a printable marksheet for every student.

Objectives



Develop a marksheet system using Python lists/dictionaries for efficient storage of student records and subject-wise marks.



Automatically calculate total marks and percentage for every student across all subjects.



Assign grades to each student based on defined percentage thresholds.



Rank students according to total marks, from highest to lowest.



Generate a clear, printable marksheet for every student, ready for distribution.

03 · LOGIC FLOW

Algorithm

01

Initialise

Set up the student records list, marks arrays, and an empty ranked list.

02

Enter Marks

Read roll number, name, and subject-wise marks for each student.

03

Calculate Total & %

Sum subject marks to get the total; compute percentage against maximum marks.

04

Assign Grade

Map the percentage to a grade using defined thresholds (A+, A, B, C, Fail).

05

Rank Students

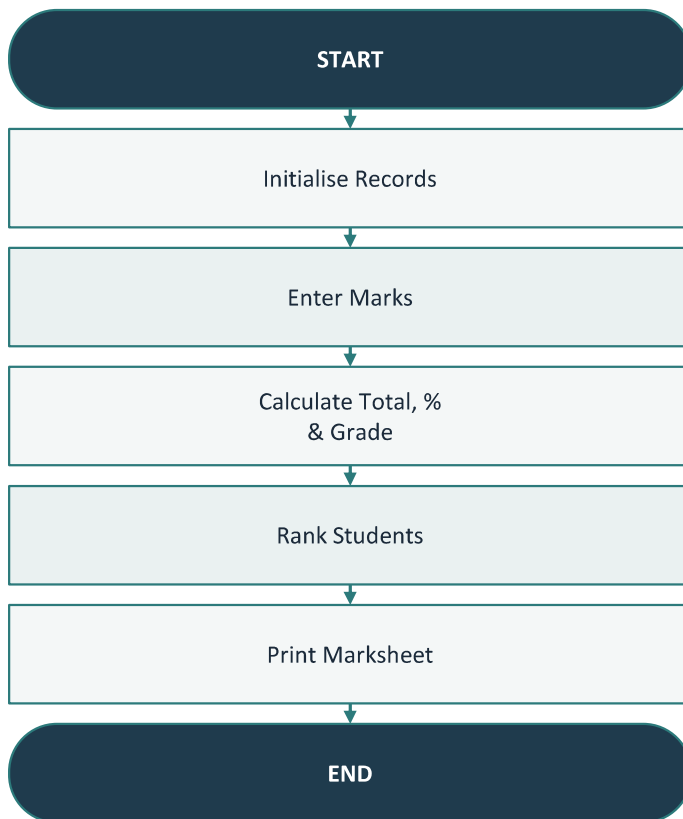
Sort all students by total marks in descending order and assign ranks.

06

Generate Marksheet

Display / print a formatted marksheet for every student, then repeat until done.

Flowchart / Pseudocode



PSEUDOCODE

```
students = []
```

```
FOR each student:
```

```
    read roll_no, name, marks[]
```

```
    total = sum(marks)
```

```
    percentage = total / max_marks * 100
```

```
    grade = assign_grade(percentage)
```

```
    students.append({...})
```

```
SORT students BY total DESCENDING
```

```
ASSIGN rank to each student
```

```
FOR each student in ranked list:
```

```
    PRINT marksheet(name, roll_no,
```

```
        marks, total, percentage,
```

```
        grade, rank)
```

Python Logic

```
subjects = ["Maths", "Physics", "Python"]
marks = [88, 76, 95]

total = sum(marks)
percentage = total / (len(marks) * 100) * 100

def assign_grade(pct):
    if pct >= 90: return "A+"
    elif pct >= 75: return "A"
    elif pct >= 60: return "B"
    elif pct >= 40: return "C"
    else: return "Fail"

grade = assign_grade(percentage)

print("Total:", total)
print("Percentage:", round(percentage, 2))
print("Grade:", grade)
```

Lists / Dicts

Store each student's roll number, name, and subject-wise marks using indexed data structures.

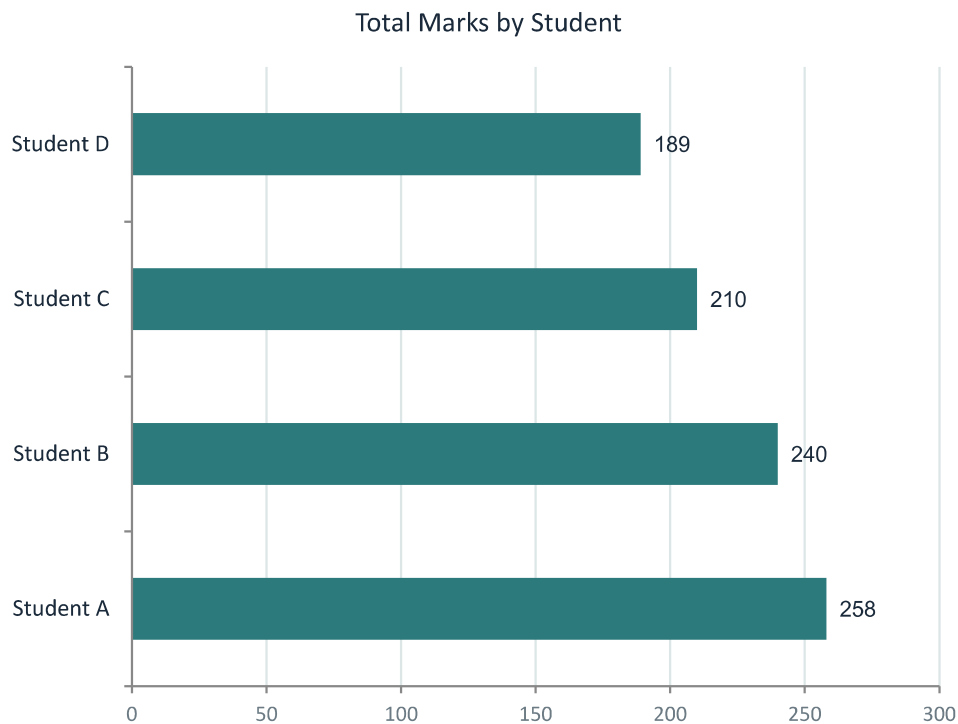
Grade Assignment

if / elif thresholds on percentage map each student to A+, A, B, C, or Fail.

Ranking

sorted() on total marks (descending) produces the class rank list.

Working & Graphical Representation



1

Calculate Totals

Each student's subject marks are summed and converted into an accurate percentage.

2

Assign Grades

Percentage thresholds automatically map to a letter grade for every student.

3

Rank Students

Students are sorted by total marks to produce the final class ranking.

4

Efficiency

$O(n)$ total/percentage calculation and $O(n \log n)$ ranking — efficient for large batches.

GitHub & Individual Contribution

GitHub Setup

Created a GitHub account and repository "Student-Marksheet-Automation", uploaded all Python files with meaningful, incremental commits. Used branches and pull requests for collaboration, tracking every project update.

My Contribution

- Problem analysis & requirement gathering
- Algorithm design for grading and ranking
- Python implementation — totals, percentage, grade assignment, ranking logic
- Testing across multiple student batches
- GitHub commits & version control
- Documentation and report preparation

Result / Expected Output

SAMPLE CONSOLE OUTPUT

Roll No: 21

Name: Aarav Sharma

Maths : 88

Physics : 76

Python : 95

Total : 259 / 300

Percentage : 86.33%

Grade : A

Rank : 2

Marksheet generated successfully.



Marks Calculated

Total and percentage computed accurately for every student.



Grade Assigned

Correct letter grade applied based on percentage thresholds.



Class Ranked

Students sorted and ranked by total marks.



Marksheet Printed

Clean, formatted marksheet ready for distribution.

Thank You